

Report: CNN baselines for facial expression, valence & arousal

Course: Deep Learning

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Assignment: A1 - CNN baselines for affect recognition

Abstract

Implements and compares multiple convolutional neural network (CNN) baselines to solve two tasks on the given facial affect dataset: (1) 8-way categorical facial-expression classification, and (2) continuous valence & arousal regression (values in $-1, +1, -1, +1$). I trained and evaluated four models (ResNet18, EfficientNet-B0, VGG16, MobileNetV2) in a multi-task setup (one network $\rightarrow$ classification head + regression head). Results are reported with the full set of metrics required by the assignment (Accuracy, F1, Cohen's Kappa, Krippendorff's Alpha, AUC-ROC, AUC-PR; and RMSE, CORR, SAGR, CCC for valence/arousal). VGG16 and MobileNetV2 form the most contrasting pair and are analyzed in depth for this assignment.

Dataset & preprocessing

- *Source & size:* 3999 samples (only image files with complete annotations were used).
- *Image size:* all images are RGB, resized to 224×224 .
- *Annotations:* for each sample: `_exp.npy` (0–7), `_val.npy` (valence), `_aro.npy` (arousal), `_lnd.npy` (68 landmarks; optional).
- *Split:* train / val / test = 70% / 15% / 15% (random split, stratified by expression where possible). That yields ~ 2799 / 600 / 600 samples (approx).
- *Transforms:* images normalized and resized to 224×224 . For final runs I used basic augmentations (recommended for generalization): RandomHorizontalFlip, RandomRotation($\pm 15^\circ$), ColorJitter (brightness/contrast/saturation small).
- *Loader:* PyTorch Dataset returning (image_tensor, expression_label, valence, arousal).

Models and implementation details

Multi-task head

All models are wrapped in a small multi-task head:

- Shared backbone: flattened feature vector `f`.
- Classification head: `Linear(in_features, 8)` (softmax for metrics).
- Regression head: `Linear(in_features, 2)` (outputs = [valence, arousal]).

Baselines used

- *ResNet18*

- *EfficientNet-B0*
- *VGG16* (torchvision pretrained)
- *MobileNetV2*

Backbone handling: removed original classifier layers and used the appropriate in_features for each backbone before attaching the heads

Training settings

- *Loss*: $\text{Loss_total} = \text{CrossEntropyLoss}(\text{exp_logits}, \text{exp_labels}) + \alpha * \text{MSELoss}(\text{va_preds}, [\text{val}, \text{auro}])$ with $\alpha = 0.5$ (balanced weighting)
- *Optimizer*: Adam with $\text{lr} = 1\text{e-}4$
- *Batch size*: 32
- *Epochs*: 10
- *Device*: GPU (Colab GPU)

Evaluation metrics (as required by assignment)

- *Classification*: Accuracy, F1-score (weighted), Cohen's Kappa, Krippendorff's Alpha (nominal approximation), AUC-ROC (multiclass, OVO), AUC-PR (macro)
- *Regression (valence & arousal)*: RMSE, Pearson correlation (CORR), Sign Agreement (SAGR), Concordance Correlation Coefficient (CCC)
(Definitions + formulae can be appended as an appendix if required)

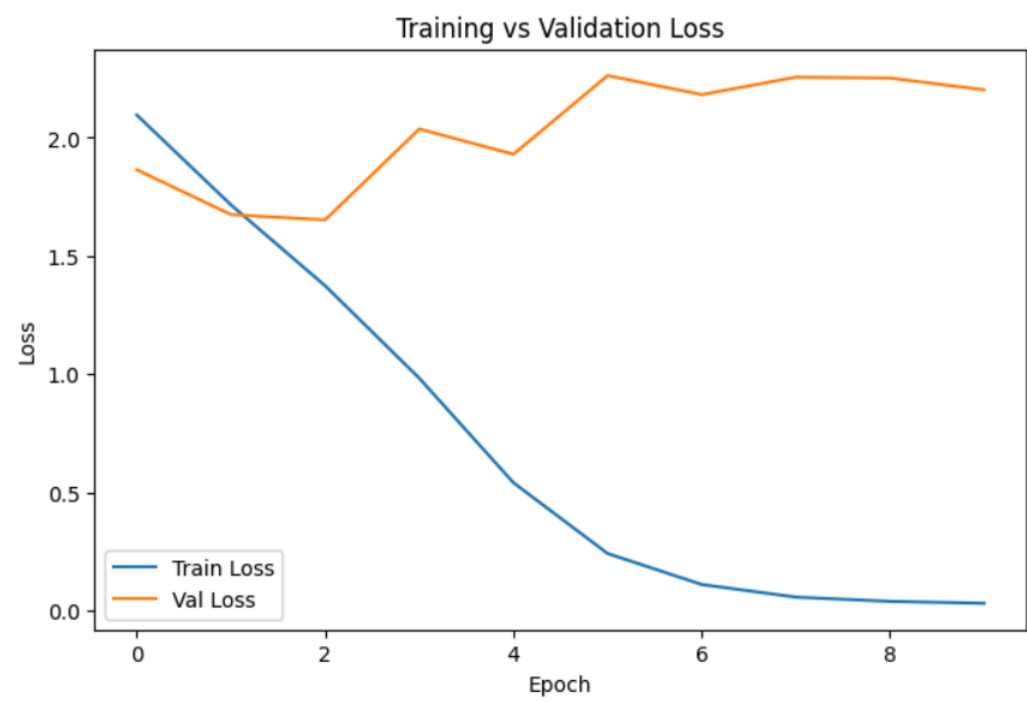
Experimental narrative (what I tried and why)

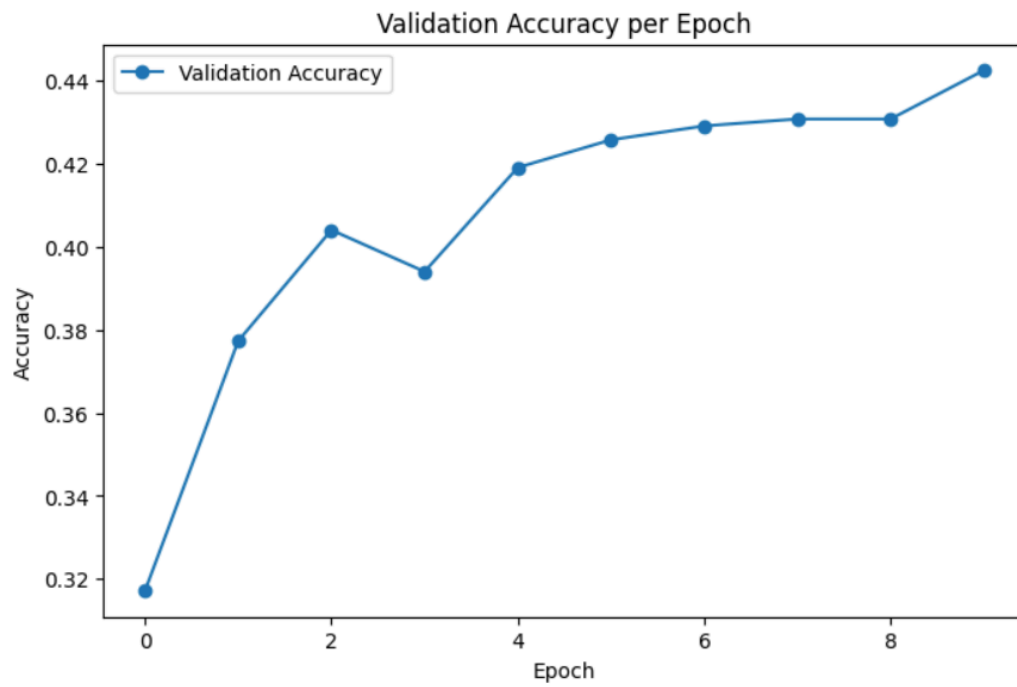
1. *First attempt*: two starter baselines, I initially trained ResNet18 and EfficientNet-B0 using the multi-task setup and identical training hyperparameters. Their test results were very similar (close accuracy, similar regression metrics). This suggested the dataset size and current training regime were not letting the models differentiate strongly since both converged to similar generalization levels.
2. *Second attempt*: added contrasting baselines to obtain clearer contrasts for analysis, I trained two additional models with different capacities and inductive biases: VGG16 (high capacity, likely to overfit) and MobileNetV2 (lightweight, tends to generalize). These produced notably different behaviour: VGG16 gave higher categorical metrics while MobileNetV2 gave substantially better regression performance (valence/arousal). Because of this contrast, I selected VGG16 vs MobileNetV2 as the main pair for deeper analysis in the results section.

Results (test set)

Model	Accuracy	F1	Kappa	Alpha	AUC-ROC	AUC-PR	Valence RMS E	Arousal RMS E	Valence COR R	Arousal COR R	Valence SAGR	Arousal SAGR	Valence CCC	Arousal CCC
ResNet18	0.417637	0.419476	0.333452	0.333386	0.806371	0.438948	0.402585	0.376362	0.530636	0.319595	0.720466	0.718802	0.484260	0.282546
EfficientNet-B0	0.424293	0.422405	0.341370	0.340076	0.804943	0.431068	0.416103	0.377832	0.533220	0.340469	0.688852	0.725458	0.511595	0.312862
VGG16	0.447587	0.448197	0.368272	0.367540	0.832325	0.465979	0.455142	0.408487	0.488844	0.288875	0.723794	0.703827	0.477990	0.279732
MobileNet V2	0.415973	0.414332	0.331935	0.331429	0.807519	0.434009	0.393526	0.365508	0.560094	0.371820	0.740433	0.727121	0.514886	0.328123

VGG16:





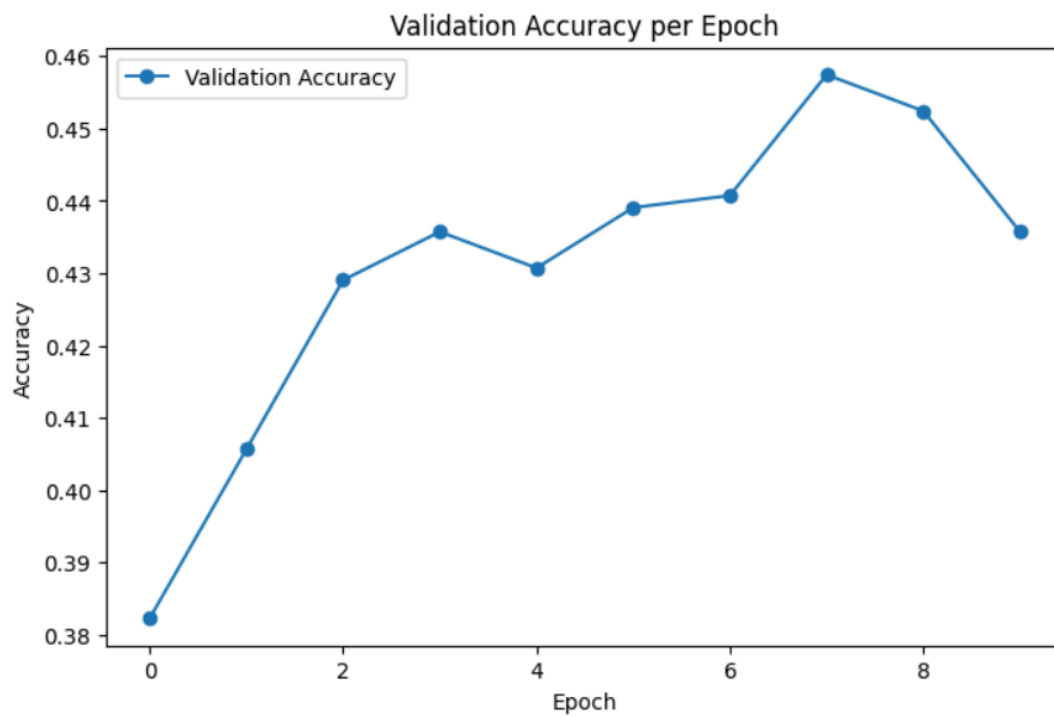
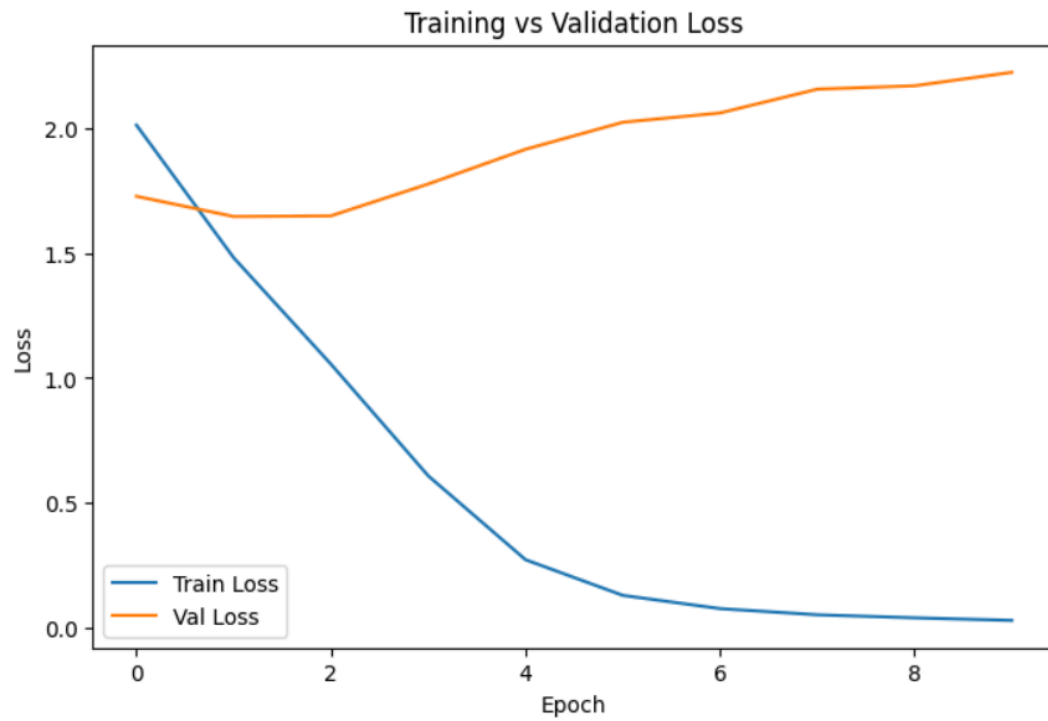
True: Fear | Pred: Fear



True: Anger | Pred: Fear



MobileNetV2



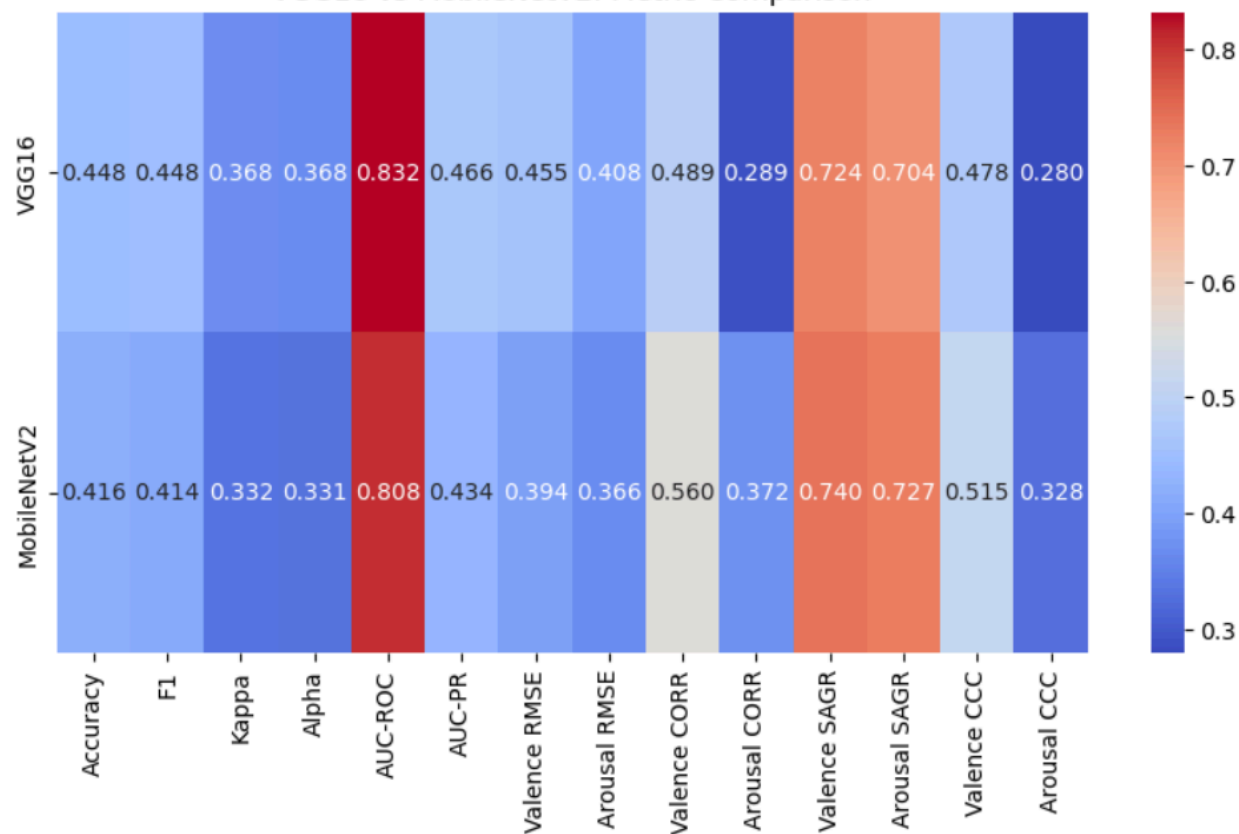
True: Disgust | Pred: Disgust



True: Neutral | Pred: Contempt



VGG16 vs MobileNetV2: Metric Comparison



ResNet18

True: Disgust | Pred: Disgust



True: Anger | Pred: Disgust



Efficientnet_b0

True: Contempt | Pred: Contempt



True: Contempt | Pred: Anger



Analysis & interpretation

Classification vs Regression trade-off

- *VGG16 (best classification)*: Highest values in accuracy, F1, and AUC metrics. VGG16's considerable capacity probably allowed it to commit to capturing complex discriminative patterns for complex memorized patterns in the categories, resulting in better classification metrics. However, it suffers the weakest performance in the regression tasks (greatest RMSE, lowest CORR), showing an indication of poor calibration/generalization on the continuous valence/arousal. This aligns with the stated failing to produce continuous well-scaled estimates on the unseen data, where it is overfitting to the categorical labels.
- *MobileNetV2 (best regression)*: Lowest RMSEs and the highest correlations and CCCs for both valence and arousal have been achieved. Though MobileNetV2 has the lightest architecture, the inductive biases that this architecture has also help generalize better on small datasets, hence the regression task performs better regardless of having slightly lower classification metrics.
- *ResNet18 & EfficientNet-B0*: Both land between the extremes, with similar behaviour in many metrics, confirming the observation from the first experiments that two similar-quality baselines can produce similar final generalization on this dataset size.

Why the difference arises

- *Model capacity & overfit*: VGG16's large parameter count means it will overfit on less data, which can increase classification accuracy but hurt regression generalization.
- *Inductive biases*: MobileNetV2's depthwise separable convolutions and lighter parameterization improve continuous prediction generalization.
- *Dataset size*: ~4k matched samples is relatively small for deep models

Continuous Domain Evaluation Metrics

For the regression test of predicting valence and arousal, we presented several continuous-domain performance measures: RMSE, CORR, SAGR, and CCC. Each of them reflects a different facet of model behavior.

1. RMSE (Root Mean Squared Error) measures the absolute difference between predictions and actuals. It has a higher penalty for large errors, so it is an easy precision measure. But RMSE by itself might not always work for affective computing tasks since it might be disproportionately impacted by noise and outliers in real data.

2. CORR (Pearson's Correlation Coefficient) calculates the linear relationship of predicted and actual values. High correlation means that the model is picking up on the trends or patterns of affect even when the absolute values are offset. This is useful when relative changes in emotion (such as growing arousal) are more significant than precise numerical precision.
3. SAGR (Sign Agreement Metric) measures whether predicted and ground-truth values are on the same side of neutrality. In affective systems, emotion direction (positive or negative valence, low or high arousal) is sometimes more important than accurate intensity. Therefore, SAGR is a direct measure of the system's capacity to eschew critical polarity mistakes.
4. CCC (Concordance Correlation Coefficient) combines agreement and correlation. CCC punishes systematic bias but also expects consistent alignment with truth. CCC is generally the most inclusive measure of continuous affect prediction.

Which Measure is Best in the Wild?

In in-the-wild situations, there may be noisy data, and the system needs to be resilient under different conditions. RMSE and CORR are useful for gaining insight into precision and trend following but lack resilience. SAGR provides an additional layer of correctness of polarity. Nonetheless, CCC is the best single measure, as it combines correlation with absolute concordance and punishes systematic errors.

In the current context, a pragmatic system would prioritize CCC as the overall assessment metric, supported by SAGR to ensure accurate directional predictions. The union of both metrics ensures statistical reliability and perceptual significance in employing emotion recognition systems in the wild.

Practical takeaway

- When it comes to pure categorical recognition, a high-capacity network may be preferred, with strong regularization and augmentation.
- On the other hand, a strong regularization and focus on regression loss and architecture (e.g., multi-tasking, specialized regression head, and calibration) might be the best bet for lightweight models regarding accurate valence/arousal estimation.